

than using the thoughts of the designer or manufacturer.

Manufacturer. It benefits the manufacturer as the effort involved in providing support for the product is less. Just like in the case of open source software, an ecosystem involving users and enthusiasts from around the world comes forth to share ideas, suggest improvements and even provide support for the open source hardware, through online discussion forums, etc. Therefore the manufacturer has to put in very less effort to provide support for the product, as most often the users solve their own issues by posting their queries online and getting answers from the community.

Regional players. A lot of regional players get access to the hardware and leverage this opportunity to come up with a business model. For example, if a mobile phone manufactured in the US is open sourced, a local designer

found enough reason to open up several of their designs—after all, it is an interesting way to drive mass adoption as well as to kindle the collective wisdom of the global community of experts to improve and revise the designs.

Open source hardware has seen mass adoption in multiple segments like mobile devices, set-top boxes, digital TV, Internet media tablets, gaming devices, IP net cameras, desktop PCs, computer vision, industrial automation, evaluation platforms and sensor networks.

Microcontrollers, chipsets, laptop designs and several component designs have been outsourced thus far. Not only the hardware designs but also the tools to work on these designs are available under the free/open source umbrella.

“A very important part of hardware is the internal cores that make up the integrated circuits on a board. These cores are combined together to produce the final integrated circuit devices. Opencores.

ECB_AT91, Amazon Kindle and Marvell SheevaPlug built around ARM SoC (system-on-a-chip).

Khasim explains that the big players are quite open to open sourcing. He says, “There is definitely no resistance from anyone on making an open source hardware, but there should be a need and acceptance of a solution on a broader scale. One will not simply open up a technology when there are no buyers for the same. Companies like Nokia, Texas Instruments, Intel, Neuros, OpenPandora, Openmoko and Innovate Solutions are already engaged in open source hardware. Open source hardware is not simple; it has its own issues and limitations in terms of expenses involved in raising and building up the community, promotions and offerings, designing the platforms, legal licensing, etc. An organisation that enters the fray of open source hardware has to prepare well in advance, with the right vision and roadmap for next-gen offerings.”

Texas Instruments believes in open platforms as it drives a broader degree of user innovation. The company is involved in major projects like Internet media tablets, gaming devices, OSDs, beagleboard.org, leopardboard.org, toasterboard.org and OMAP Zoom. Moving forward, it expects many low-cost, open-source hardware designs from partners and customers using TI technology.

“The products we create today have to meet the exponentially growing consumer demands like low cost, low power, high performance and features. By supporting open platforms and products we are definitely enabling the innovative engineering minds and an open ecosystem to create the next-generation products,” adds Khasim.

Contradicting the view that the big players in the industry have no reason to resist open source hardware, D’Souza remarks, “FLOSS and FLOSH tend to fragment the market and reduce the entry barriers to near zero. Consequently, big players always resist such fragmentation and try hard to push a one-size-fits-all philosophy.”

However, D’Souza explains that

Even the big players have found enough reason to open up several of their designs—after all, it is an interesting way to drive mass adoption as well as to kindle the collective wisdom of the global community of experts to improve and revise the designs

in India can manufacture the same at less cost and give local and regional support to his customer. The cost of importing the product is reduced drastically. Open sourcing builds a global ecosystem as the applications built and designed on the platform (anywhere in the world) become globally accessible.

Open sourcing paves the way to innovative usage of the product. An open design can be tuned to work for a different application. For instance, an open mobile Internet device can be used by another person to address a requirement in telemedicine.

A populated playing field

Open source hardware concept has been adopted by several players ranging from big to small. It is no longer associated only with geeks, start-ups and hobbyists. Even the big players have

org has a huge number of contributors providing FLOSH cores. The collection includes 8-, 16- and 32-bit processors, security encryption devices, codecs, memory management units, etc. These will be the foundation of tomorrow’s hardware,” adds D’Souza.

Apart from interesting community projects such as Beagle board and Leopard board, there are numerous large companies that have involved themselves in open source hardware. The list includes companies such as Texas Instruments, Nokia, Intel and Google. Large FPGA players like Xilinx, Altera and Lattice also actively support FLOSH.

Open source hardware has undoubtedly been the wind beneath the wings of several well-known projects such as the OLPC (one laptop per child), Simputer (an example closer home), Openmoko,